

ASSUMPTION UNIVERSITY
VINCENT MARY SCHOOL OF ENGINEERING, SCIENCE AND TECHNOLOGY
CSX3008 (OS) Quiz 2 (2/2025) **SOLUTION**

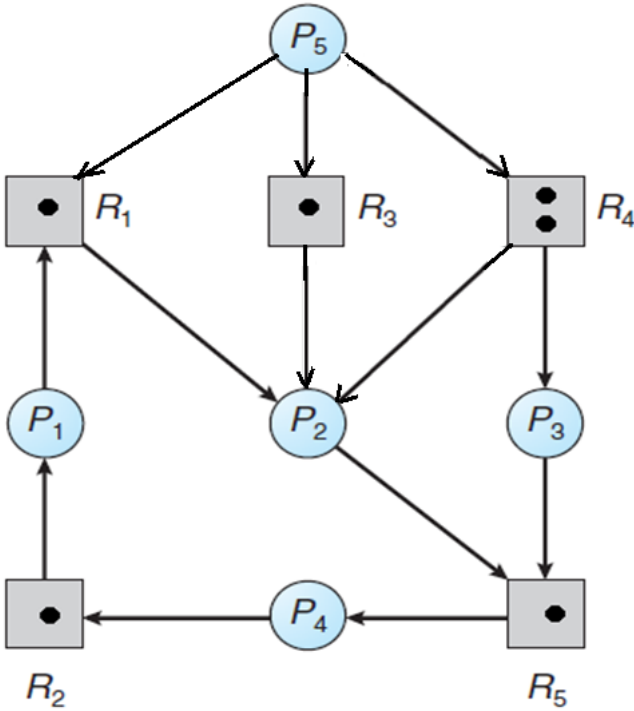
There are 20 questions, and the total is 60 points (3 hours).

Instructions:

- 1. Read the questions carefully and answer all questions on the answer sheet provided (*do not use any pencil for your answer*).
- 2. Write your **Name and ID Number** on the question paper and the answer booklet.
- 3. Students are **allowed to bring a non-programmable scientific calculator** into the exam room.
- 4. This is a **closed-book** examination.

Name: I.D. No.: Sec.

- 1. (1 point) Sometimes, a waiting process can never change its state again during a computation because other processes hold the requested resources. This situation is called a -----**deadlock**-----.
- 2. (3 points) Consider the **resource allocation graph (RAG)** shown in the **Figure** below, where R_1, R_2, R_3, R_4 , and R_5 are resources with their units, and P_1, P_2, P_3, P_4 , and P_5 are processes.
 - 2.1.(2 points) Attempt the **graph reduction** method and determine whether a **deadlock** exists.
Deadlock exists because the RAG cannot be reduced.
 - 2.2. (1 point) What is the drawback of the RAG? **A process in a RAG can request only one resource unit at a time.**



- 3. (5 points) Consider a system snapshot with its **Max** and **Allocation** resource matrices shown in the **figure** below (where P_0, P_1, P_2, P_3 , and P_4 are processes and A, B, C , and D are resources, respectively). Assume the resources have 14, 12, 7, and 8 units. If the system is safe, show its **safe state sequence**. Otherwise, indicate the **completed processes** (if any) before the deadlock.

Max					Allocation				
	A	B	C	D		A	B	C	D
P_0	6	2	3	7	P_0	3	0	1	4
P_1	3	2	1	1	P_1	2	2	1	0
P_2	3	3	2	1	P_2	3	1	2	1
P_3	4	8	5	3	P_3	0	5	1	0
P_4	8	3	4	5	P_4	4	2	1	2

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Unsafe state. The algorithm stopped at iteration 2 due to insufficient resources in the available resource vector to support P0, P3, and P4. Hence, the execution sequence of processes could not reach a safe state. However, processes P1 and P2 could complete their execution.

4. (1 point) A compiler binds the symbolic address (source code address) of a variable in a program into a data memory address (main memory address) at compile time. This is called ---- binding (choose a suitable answer from the given choices).
- a. Dynamic b. Run-time c. **Static** d. Virtual e. None
5. (1 point) Check whether the following statement is **true** or **false**: Registers that are built into each CPU core are generally accessible within one cycle of the CPU clock. **True**
6. (1 point) An address generated by the CPU for accessing instructions from main memory is referred to as ----- (find a suitable answer from the given choices).
- a. Physical address b. **Logical Address** c. I/O address d. Memory Address e. None
7. (1 point) With ----- loading, a routine or a function in a program is not loaded into main memory until it is called (find a suitable answer from the given choices).
- a. Physical b. Static c. **Dynamic** d. Virtual e. Logical
8. (1 point) The allocation and deallocation of the processes in main memory causes the creation of scattered holes, which cannot be used for processes of a large size. This issue in memory is called - ----- fragmentation. **External**
9. (6 points) Consider the **five** memory holes: 530KB, 440KB, 180KB, 600KB, and 220KB. Show how the **first-fit**, **best-fit**, and **worst-fit** dynamic hole allocation procedures support the following **five** processes: 160KB, 570KB, 450KB, 390KB, and 190KB. Rank the procedures by comparing the total hole generated after their hole allocation. **The Best-fit procedure shows better performance.**

FIRST-FIT DYNAMIC MEMORY HOLE ALLOCATION PROCEDURE				
Process	Process Size	Hole selected	Hole size	Left over space
P1	160 KB	H1	530 KB	370 KB
P2	570 KB	H4	600 KB	30 KB
P3	450 KB	-NOT-	0 KB	0 KB
P4	390 KB	H2	440 KB	50 KB
P5	190 KB	H1	370 KB	180 KB

The following Holes remains after the FIRST-FIT procedure:
{180KB, 50KB, 180KB, 30KB, 220KB} ==> Total 660KB.

BEST-FIT DYNAMIC MEMORY HOLE ALLOCATION PROCEDURE				
Process	Process Size	Hole selected	Hole size	Left over space
P1	160 KB	H3	180 KB	20 KB
P2	570 KB	H4	600 KB	30 KB
P3	450 KB	H1	530 KB	80 KB
P4	390 KB	H2	440 KB	50 KB
P5	190 KB	H5	220 KB	30 KB

The following Holes remains after the BEST-FIT procedure:
{20KB, 30KB, 30KB, 50KB, 80KB} ==> Total 210KB.

WORST-FIT DYNAMIC MEMORY HOLE ALLOCATION PROCEDURE				
Process	Process Size	Hole selected	Hole size	Left over space
P1	160 KB	H2	600 KB	440 KB
P2	570 KB	-NOT-	0 KB	0 KB
P3	450 KB	H3	530 KB	80 KB
P4	390 KB	H2	440 KB	50 KB
P5	190 KB	H5	440 KB	250 KB

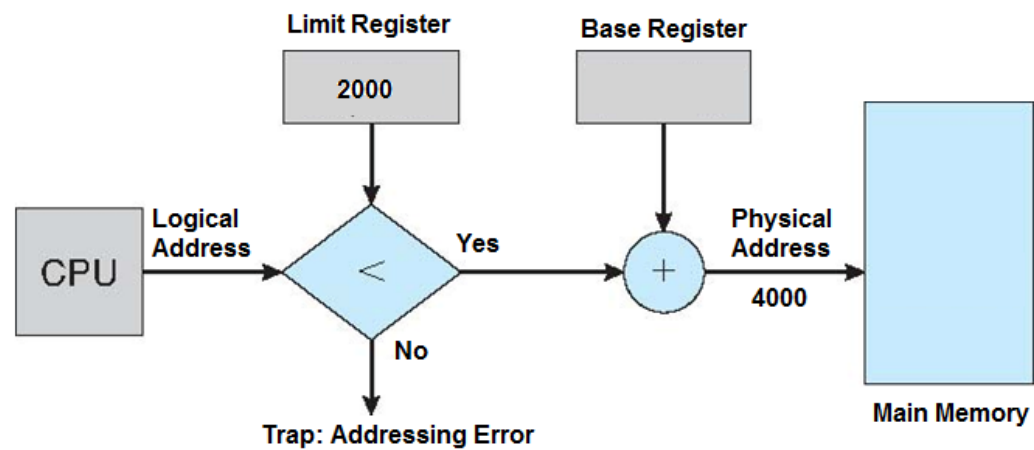
The following Holes remains after the WORST-FIT procedure:
{250KB, 220KB, 180KB, 80KB, 50KB} ==> Total 780KB.

Conclusion:

The BEST-FIT Procedure is the suitable one for the given Processes.

10. (7 points) An OS's basic memory protection scheme is shown in the **Figure** below. Answer the following based on the figure:
- 10.1. (2 points) What is the *range of the logical address*? **0-1999**
- 10.2. (1 point) How many instructions are in the process (each logical address indicates an instruction in the instruction memory)? **2000**
- 10.3. (1 point) Show a logical address that can cause a trap. **2000 or above**
- 10.4. (1 point) Find the values of the **Base register** if the logical address is 1000. **3000**
- 10.5. (2 points) What is the *physical address range* of the process (use the idea from **10.4**) **3000-4999**

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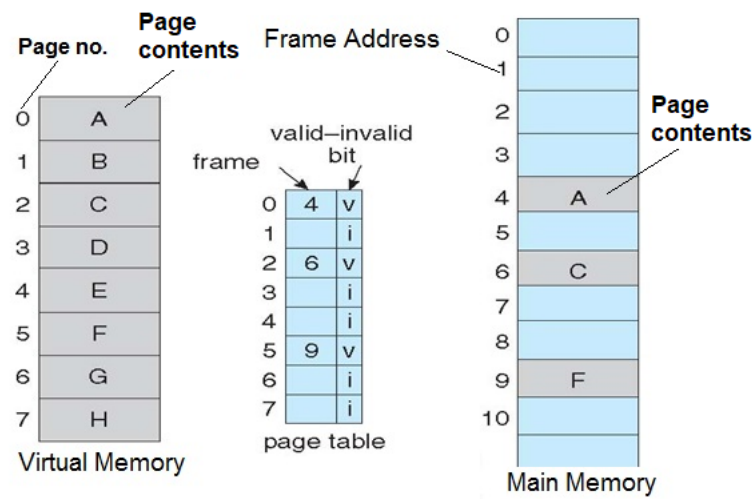


11. (2 points) Consider a computer with a 32-bit virtual address (VA) and a 4k-bit page size. How many **rows** are in the direct page table? **$2^{32}/2^{12} = 2^{20} = 1\text{M}$ rows in the page table.**
12. (3 points) A section of the **page table** is shown in the **Figure** below. Based on the page table, show the **physical addresses** (in *decimal*) of the following **virtual addresses**:
- 12.1. 311_{16} **149** 12.2. $E07_{16}$ **103** 12.3. $F8C_{16}$ **284**

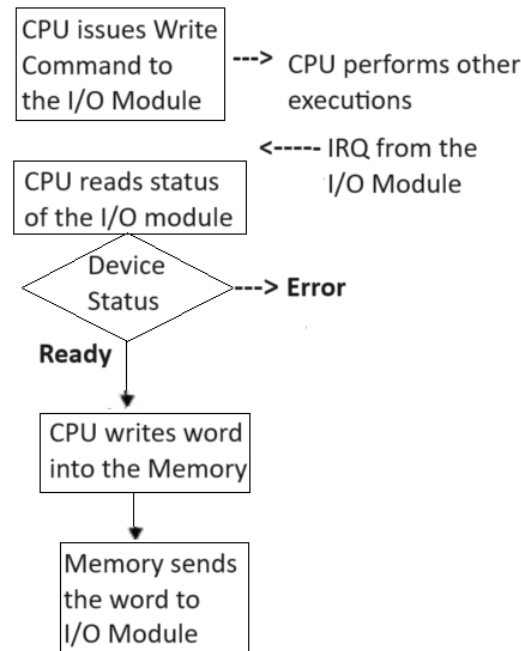
Page No.	v/i bit	Page Frame
0	v	A
1	v	7
2	v	0
3	v	B
---	-	--
E	v	8
F	v	C

13. (8 points) A direct virtual memory (VM) system is shown in the **Figure** below. Answer the following based on the VM:
- 13.1. (1 point) What is the relationship between the page table and the main memory? **The page table lists the page frame numbers of the pages loaded into memory.**
- 13.2. (1 point) What does the *valid-invalid* (v/i) bit mean in the page table? **The v/i bit indicates whether a page is in memory and helps identify a page fault.**
- 13.3. (2 points) Assume **page no. 4** (page E) is selected for the subsequent execution. Briefly describe how the OS handles the situation. **The situation is a page fault, and the OS PFHR identifies the missed page from the VM, loads it into an available memory frame, and finally updates the page table.**
- 13.4. (2 points) Is any **page replacement** algorithm needed to solve the issue in (13.2)? Why? **In the VM scenario, there is no need to execute a page replacement algorithm for the missed page because there are free frames in main memory.**
- 13.5. (2 points) Suppose there are **1M** rows in the page table. How many pages are in the VM? **1M**

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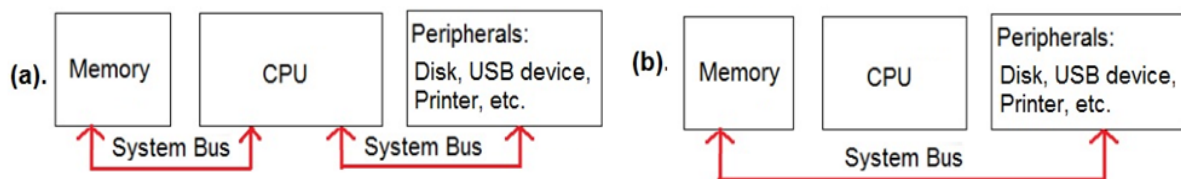
14. (6 points) Assume a user process is executing and has **five pages**; its page reference string is “213425125312513”. There are **three memory frames** in the system for page allocation. Show the performance of **optimal**, **FIFO**, and **LRU** page replacement algorithms based on their number of **page fault** resolutions (show your *page allocation for each page fault resolution* by the algorithms).
Optimal: (3 ← 4) PF, (4 ← 5) PF, (5 ← 3) PF, (2 ← 5) PF = Total 4 PFs resolved with page replacement.
LRU: (2 ← 4) PF, (1 ← 2) PF, (3 ← 5) PF, (4 ← 1)PF, (1 ← 3) PF, (2 ← 1) PF, (5 ← 2) PF, (3 ← 5)PF, (2 ← 3)PF = Total 9 PFs resolved with page replacement.
FIFO: (2 ← 4) PF, (1 ← 2) PF, (3 ← 5) PF, (4 ← 1)PF, (2 ← 3) PF, (5 ← 2) PF, (1 ← 5) PF, (3 ← 1)PF, (2 ← 3)PF = Total 9 PFs resolved with page replacement.
15. (1 point) The installed ----- helps the OS identify an I/O device during a system call by the user application (find a suitable answer from the given choices).
a. Device controller b. **Device driver** c. Device request d. Device name e. None
16. (3 points) Identify the nature of the **I/O operation** shown in the **Figure** below. Why?



This is an **Interrupt-driven I/O operation** because the **CPU does not need to wait for the I/O device's status after issuing an I/O command**. Then the CPU can continue executing until the corresponding I/O controller interrupts it by asserting an **IRQ (interrupt request)** on one of the CPU's IRQ pins to indicate the device status (**Ready or Error**), the CPU stops its current execution and handles the I/O operation by the OS's interrupt service routine (ISR) for the I/O device

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17. (1 point) Check whether the following statement is **true** or **false**: A system call is a software interrupt that a user application uses to initiate an I/O operation. **True**
18. (1 point) Check whether the following statement is **true** or **false**: The I/O controller request is a hardware interrupt to the CPU that results from a system call. **true**
19. (1 point) Check whether the following statement is **true** or **false**: A page fault is a *user-maskable* interrupt. **False**
20. (7 points) Two types of I/O operations are shown in **Figures (a) and (b)**. Based on the figure, answer the following:



- 20.1. (1 point) Identify the nature of the I/O operation in **Figure (a)**. **Programmed I/O or direct I/O operation.**
- 20.2. (1 point) Identify the nature of the I/O operation in **Figure (b)**. **Direct Memory Access (DMA) I/O operation.**
- 20.3. (5 points) Which one is suitable for *large data transfer*? Why? Briefly describe its operation steps.

Once the system call causes a DMA I/O operation (Figure b):

- a). The DMA controller collects the status of its I/O device and then interrupts the CPU as **DRQ** (DMA Request).
 - b). If the CPU accepts the DRQ from the DMA controller, it sends a **DACK** (DMA Acknowledgement) signal back to the DMA controller.
 - c). It causes the DMA controller to seize or borrow (called **cycle stealing**) the *system bus* (*address bus, data bus, and control bus*) from the CPU.
 - d) Then, the DMA controller places the desired memory address on the address bus for data transfer between the **DMA I/O device** and the main memory without CPU intervention.
- The DMA controller cancels the **DRQ** signal, and the CPU cancels the **DACK** signal to indicate the end of the DMA operation.

=====End of Quiz=====